

NUS QF3101: Investment Instrument and Risk Management

March 22, 2023

Computing Assignment

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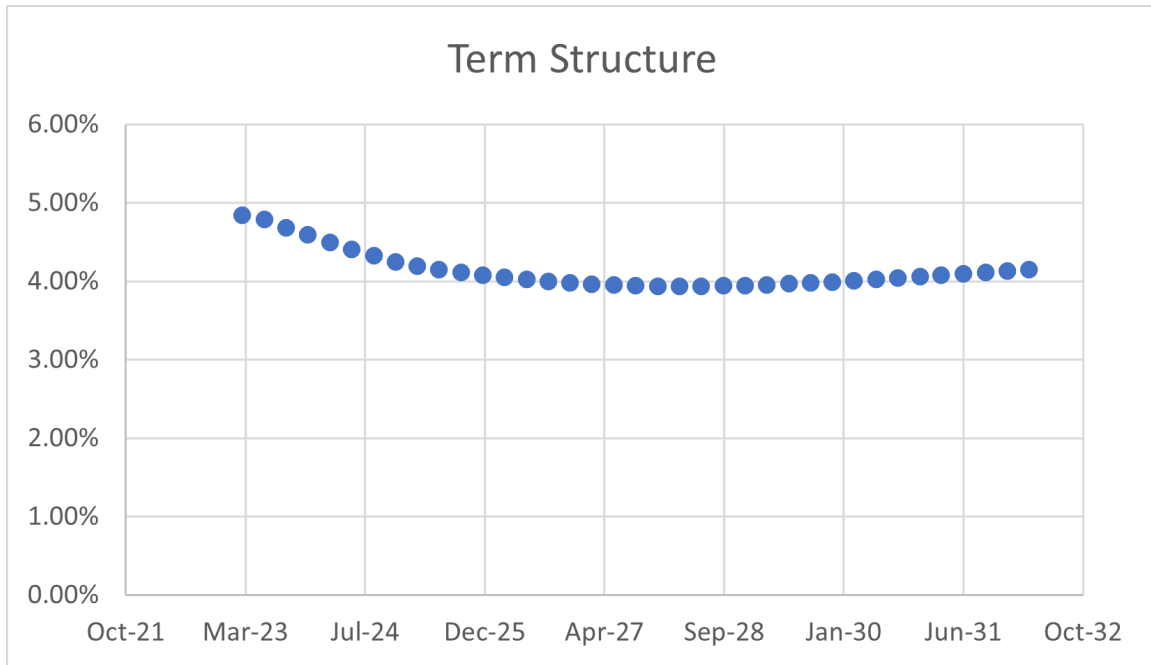
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1 Three-month SOFR Futures and Deferred IRS

1.a Data Collection and Preparation

THREE-MONTH SOFR FUTURES - QUOTES										VENUE: GLOBEX											
MONTH	OPTIONS	CHART	LAST	CHANGE	PRICE OFFER	OPEN	HIGH	LOW	VOLUME	UPDATED	MONTH	OPTIONS	CHART	LAST	CHANGE	PRICE OFFER	OPEN	HIGH	LOW	VOLUME	UPDATED
JAN 2023 SO237	☑	⬆	-	-	95.3925	-	-	-	0	22:13:12 CT 14 Mar 2023	MAR 2027 SO247	☑	⬆	95.78	-0.85 (-0.85%)	95.83	95.745	95.885	95.74	5,515	01:02:06 CT 15 Mar 2023
FEB 2023 SO238	☑	⬆	-	-	95.29	-	-	-	0	22:13:12 CT 14 Mar 2023	JUN 2027 SO287	☑	⬆	95.78	-0.845 (-0.85%)	95.825	95.755	95.885	95.74	5,319	01:28:58 CT 15 Mar 2023
MAR 2023 SO239	☑	⬆	95.095	-0.865 (-0.87%)	95.16	95.1875	95.1225	95.895	21,277	01:19:35 CT 15 Mar 2023	SEP 2027 SO337	☑	⬆	95.74	-0.84 (-0.84%)	95.78	95.785	95.77	95.78	6,773	01:29:57 CT 15 Mar 2023
APR 2023 SO240	☑	⬆	-	-	95.195	-	-	-	0	22:13:12 CT 14 Mar 2023	DEC 2027 SO327	☑	⬆	95.785	-0.84 (-0.84%)	95.785	95.87	95.73	95.66	5,234	01:21:08 CT 15 Mar 2023
MAY 2023 SO243	☑	⬆	-	-	95.285	-	-	-	0	22:13:12 CT 14 Mar 2023	MAR 2028 SO298	☑	⬆	95.785	-0.84 (-0.84%)	95.785	95.685	95.725	95.66	3,385	01:29:57 CT 15 Mar 2023
JUN 2023 SO246	☑	⬆	95.175	-0.74 (-0.75%)	95.335	95.355	95.285	95.175	53,542	01:22:15 CT 15 Mar 2023	SEP 2028 SO308	☑	⬆	-	-	95.715	-	-	-	9	00:46:55 CT 15 Mar 2023
SEP 2023 SO253	☑	⬆	95.41	-0.225 (-0.34%)	95.635	95.685	95.685	95.485	41,428	01:22:15 CT 15 Mar 2023	DEC 2028 SO328	☑	⬆	-	-	95.67	-	-	-	7	00:46:55 CT 15 Mar 2023
DEC 2023 SO255	☑	⬆	95.585	-0.25 (-0.28%)	95.835	95.795	95.795	95.585	38,578	01:22:15 CT 15 Mar 2023	MAR 2029 SO338	☑	⬆	-	-	95.645	-	-	-	0	22:13:12 CT 14 Mar 2023
MAR 2024 SO264	☑	⬆	95.87	-0.285 (-0.21%)	95.875	95.82	95.84	95.87	29,349	01:22:15 CT 15 Mar 2023	JUN 2029 SO348	☑	⬆	-	-	95.615	-	-	-	0	22:13:12 CT 14 Mar 2023
JUN 2024 SO269	☑	⬆	95.18	-0.165 (-0.17%)	95.285	95.22	95.225	95.49	25,888	01:22:15 CT 15 Mar 2023	SEP 2029 SO353	☑	⬆	-	-	95.59	-	-	-	0	22:13:12 CT 14 Mar 2023
SEP 2024 SO274	☑	⬆	95.585	-0.135 (-0.14%)	95.44	95.40	95.415	95.38	28,759	01:22:15 CT 15 Mar 2023	DEC 2029 SO358	☑	⬆	-	-	95.565	-	-	-	0	18:26:47 CT 14 Mar 2023
DEC 2024 SO284	☑	⬆	95.43	-0.185 (-0.11%)	95.535	95.50	95.585	95.42	17,733	01:22:15 CT 15 Mar 2023	MAR 2030 SO368	☑	⬆	-	-	95.55	-	-	-	0	18:27:08 CT 14 Mar 2023
MAR 2025 SO293	☑	⬆	95.495	-0.48 (-0.68%)	95.575	95.55	95.555	95.485	18,147	01:22:16 CT 15 Mar 2023	JUN 2030 SO369	☑	⬆	-	-	95.53	-	-	-	0	22:13:12 CT 14 Mar 2023
JUN 2025 SO295	☑	⬆	95.52	-0.865 (-0.87%)	95.585	95.55	95.565	95.51	7,272	01:22:15 CT 15 Mar 2023	SEP 2030 SO378	☑	⬆	-	-	95.52	-	-	-	0	22:13:12 CT 14 Mar 2023
SEP 2025 SO295	☑	⬆	95.555	-0.46 (-0.86%)	95.615	95.56	95.595	95.54	7,415	01:22:15 CT 15 Mar 2023	DEC 2030 SO329	☑	⬆	-	-	95.51	-	-	-	0	22:13:12 CT 14 Mar 2023
DEC 2025 SO295	☑	⬆	95.62	-0.46 (-0.86%)	95.68	95.61	95.65	95.68	6,587	01:21:43 CT 15 Mar 2023	MAR 2031 SO381	☑	⬆	-	-	95.515	-	-	-	0	18:28:35 CT 14 Mar 2023
MAR 2026 SO296	☑	⬆	95.655	-0.46 (-0.86%)	95.715	95.635	95.68	95.63	6,274	01:22:16 CT 15 Mar 2023	JUN 2031 SO381	☑	⬆	-	-	95.58	-	-	-	0	18:27:12 CT 14 Mar 2023
JUN 2026 SO296	☑	⬆	95.685	-0.46 (-0.86%)	95.745	95.655	95.71	95.65	3,348	01:22:08 CT 15 Mar 2023	SEP 2031 SO381	☑	⬆	-	-	95.485	-	-	-	0	22:13:12 CT 14 Mar 2023
SEP 2026 SO296	☑	⬆	95.715	-0.45 (-0.85%)	95.765	95.685	95.74	95.675	3,346	01:22:08 CT 15 Mar 2023	DEC 2031 SO321	☑	⬆	-	-	95.475	-	-	-	0	18:27:19 CT 14 Mar 2023
DEC 2026 SO295	☑	⬆	95.76	-0.445 (-0.85%)	95.885	95.725	95.78	95.72	4,883	01:22:08 CT 15 Mar 2023	MAR 2032 SO382	☑	⬆	-	-	95.465	-	-	-	0	18:27:43 CT 14 Mar 2023
MAR 2027	—									01:22:08 CT	JUN 2032 SO382	☑	⬆	-	-	95.455	-	-	-	0	18:26:47 CT 14 Mar 2023

1.b Implied Forward SOFR Term Structure



The term structure is downward sloping for at least the next 4 years, which signifies that the economy is in, or about to enter, a recessive period. There is a slight uptrend afterwards till the end of the decade, signifying a predicted expansionary period.

1.c Swap Rate for Deferred Interest Rate Swap

K	1	2	3	4	5	6	7	8
Jun-23	4.36%	3.99%	3.80%	3.67%	3.61%	3.57%	3.56%	3.56%
Sep-23	4.12%	3.82%	3.68%	3.57%	3.53%	3.51%	3.51%	3.51%
Dec-23	3.91%	3.70%	3.58%	3.50%	3.47%	3.47%	3.47%	3.48%
Mar-24	3.73%	3.59%	3.49%	3.44%	3.43%	3.43%	3.44%	3.46%

Similar to the term structure, the swap rate drops for longer swap periods. It is also lower for further deferred start periods compared to earlier.

2 Efficient Equity Portfolios

2.a Data collection and preparation.

Average rate of return

$$r = \begin{bmatrix} 0.5583\% & 0.1963\% & 0.1510\% & 0.2355\% & -0.0618\% \end{bmatrix}^T$$

Covariance matrix

$$C = \begin{bmatrix} 0.00176 & 0.00064 & 0.00061 & 0.00099 & 0.00074 \\ 0.00064 & 0.00094 & 0.00070 & 0.00111 & 0.00080 \\ 0.00061 & 0.00070 & 0.00179 & 0.00100 & 0.00075 \\ 0.00099 & 0.00111 & 0.00100 & 0.00392 & 0.00179 \\ 0.00074 & 0.00080 & 0.00075 & 0.00179 & 0.00297 \end{bmatrix}$$

On average, AAPL performs best, while HAS performs worst. The covariance matrix is entirely positive, so it indicates that all 5 stocks tend to move in the same direction.

2.b Minimum Variance Point Portfolio And An Efficient Portfolio.

Minimum Variance portfolio with weight vector

$$w_G = \begin{bmatrix} 0.19281 & 0.69828 & 0.15050 & -0.10442 & 0.06284 \end{bmatrix}^T$$

with expected rate of return

$$r_G = 0.2390\%$$

and variance 0.000818.

An efficient portfolio with weight vector

$$w_m = \begin{bmatrix} 1.159769 & 0.311124 & -0.061948 & 0.049304 & -0.458249 \end{bmatrix}^T$$

with expected rate of return

$$r_m = 0.7392\%$$

and variance 0.002531.

It is clear that any portfolio other than the minimum variance portfolio will have a higher variance.